Crescent School

Periodic Trends Worksheet

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1. a) In your own words, explain atomic radius.
   b) Describe and explain the periodic trends in atomic radius.
   c) Rank the following elements by increasing radius: C, Al, O, K

2. a) In your own words, explain first ionization energy. Include a chemical equation for the element X, in your answer. Include physical states in your equation.
   b) Why are IE values endothermic?
   c) Describe and explain the general periodic trend in ionization energy across a period left to right, and down a group.

3. In general, what is the relationship between the periodic trends for atomic radius and ionization energy?

4. a) Consulting only a Periodic Table—not IE values, circle the atom in each pair that has a higher ionization energy: Li & Be; Ca & Br; Na & K; P & Ar; Cl or Si.
   b) Explain why F has a higher 1^{st} IE than I.

5. Consider the following data for the successive ionization of Al:

First IE:	**578 kJ/mol**
Second IE:	**1820 kJ/mol**
Third IE:	**2750 kJ/mol**
Fourth IE:	**11,600 kJ/mol**

   a) Explain the approximately steady increase in the first three IE values.
   b) Explain why the fourth IE value is much higher than predicted by the trend in the first three values. Include the electron configuration in your answer.

6. a) In your own words, explain electron affinity and electronegativity.
   b) Describe the periodic trends of each of the above.
   c) Explain why electron affinity values are generally exothermic. What does this say about the stability of species $X^{-}(g)$ relative to $X(g)$?
   d) In the light of your answer to part (c) above, explain why the noble gases have endothermic electron affinity values.
7. In general, what is the relationship between the periodic trends in electron affinity and electronegativity? Explain why this relationship makes sense.

8. Explain why there are no EN values for noble gasses.

9. Using only a periodic table, and not a set of EN values, circle the atom in each pair with the greater EN value: Ca & Ga; Br & As; Li & O, Ba & S; Cl & S; O & S

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